

**CMR COLLEGE OF ENGINEERING & TECHNOLOGY**
(UGC AUTONOMOUS)

B.Tech II Semester Supplementary Examinations March/April-2023

Course Name: ODES & MULTIVARIABLE CALCULUS**(Common for all Branches)****Date: 21.03.2023 AN****Time: 3 hours****Max.Marks: 70****(Note: Assume suitable data if necessary)****PART-A****Answer all TEN questions (Compulsory)****Each question carries TWO marks.****10x2=20M**

1. Define an Integrating factor. 2 M
2. State Newton's law of cooling. 2 M
3. Find the complementary function of $(D^3 - D)y = x$ 2 M
4. Find the particular integral of $(D^2 + a^2)y = \cos ax$ 2 M
5. Evaluate $\int_0^1 \int_0^x e^x dx dy$ 2 M
6. Change the order of integration of $\int_0^1 \int_x^1 dx dy$ 2 M
7. If $\phi = x^2 + y^2 + z^2 - 3xyz$ then find $\text{grad } \phi$ 2 M
8. If $\vec{F} = yz \vec{i} + zx \vec{j} + xy \vec{k}$ then find $\text{curl } \vec{F}$ 2 M
9. Define circulation. 2 M
10. State Green's theorem in a plane. 2 M

PART-B**Answer the following. Each question carries TEN Marks.****5x10=50M**

- 11.A). i) Solve $x \log x \frac{dy}{dx} + y = 2 \log x$ 5M
- ii) The number N of bacteria in a culture grew at a rate proportional to N. The value of N was initially 100 and increased to 332 in one hour. What was the value of N after $1\frac{1}{2}$ hour? 5M

OR

11. B). i) Solve $2px = y - y^2 p^3$ 5M
- ii) Solve $(2x + e^y) dx + x e^y dy = 0$ 5M

12. A). Solve $(D^2 - 2D + 1)y = x^2 e^{3x} - \sin 2x + 3$ 10M

OR

12. B). Using method of variation of parameters, Solve $(D^2 + 4)y = \tan 2x$ 10M

13. A). Using polar coordinates, evaluate $\int_0^a \int_0^{\sqrt{a^2 - x^2}} y \sqrt{x^2 + y^2} dx dy$ 10M

OR

13. B). Evaluate $\int_0^1 \int_0^{1-z} \int_0^{1-y-z} xyz dx dy dz$ 10M

14. A). i) Find the directional derivative of $x^2 yz + 4xz^2$ at $(1, -2, -1)$ in the direction of $2\vec{i} - \vec{j} - 2\vec{k}$ 5M
- ii) Find p so that $\vec{f} = (x + 3y)\vec{i} + (y - 2z)\vec{j} + (x + pz)\vec{k}$ is solenoidal. 5M

(P.T.O..)

OR

14. B). Show that $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ is Irrotational hence find its scalar potential function. 10M

15. A). Evaluate $\int_S \vec{F} \cdot \vec{n} \, ds$ where $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ where S is the surface of the cube bounded by $x=0, x=a, y=0, y=a, z=0, z=a$ 10M

OR

15. B). Verify Gauss divergence theorem for $\vec{F} = (x^3 - yz)\vec{i} - 2x^2 y\vec{j} + z\vec{k}$ taken over the surface of the cube bounded by the planes $x = a, y = a, z = a$ and coordinate planes 10M
